☐ CTL Provisional DECISION SHEET (PDSH)
☑ CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	**Subclause(s)**	**Tracking No.**
IEC 62930: 2017	7.1 Table 3-Ref No.10	DSH 2233
Category		
CABL		
Subject	**Keywords**	**Developed by**
Special requirements of the damp heat test	- Tensile test - Cable sample - Damp heat test	ETF 6

Question

According to clause 7.1 Table 3-Ref No.10 in IEC 62930:2017, the PV cable is required for Damp heat test, where IEC 60068-2-78 is only specified.

However, the method of the tensile test (test for determining the mechanical properties) is not detailed in IEC 60068-2-78.

Thus, there are two questions for this test.

Q1: Which standard shall be followed for the tensile test in Damp heat test?

Q2: When to prepare the test pieces for tensile test, before or after damp heat treatment?

□ Option1: apply damp heat treatment on the completed cable sample, then prepare the test pieces from the completed cable sample for tensile test.

□ Option2: prepare the test pieces from the completed cable sample first. The test pieces are subjected to the damp heat treatment, followed by tensile test.

Decision

In the Damp heat test, the tensile test (test for determining the mechanical properties) shall follow the standard IEC 60811-501. A sample of completed cable should be subjected to damp heat treatment, then the test pieces for tensile test shall be prepared from the sample of finished cable.

Explanatory notes

Enquiry:
Date: 2023-08-04

PDSH:
Year of publication: 2023
To be approved at CTL plenary: 2024

DSH
Publication date: 2024-09-16

Rationale:

1) IEC 60811-501 is the specific standard for determining the mechanical properties of insulating and sheathing compounds and shall be adopted in the Damp heat test of IEC 62930: 2017.

2) As mentioned in cl 4.4 of IEC 60068-2-78, “the specimen shall be introduced in the unpacked, switched off, ready for use state.” In other words, the specimen for damp heat treatment shall be the completed cable, instead of sheath pieces, thus Opition1 is appropriate.

Disclaimer: This document is controlled and has been released electronically.
Only the version on the IECEE Website is the current document version.